PUBUDU GUNASEKARA

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MS in Computer Science, Northeastern | Seeking Software Engineering Internship, Summer 2027

EDUCATION

Northeastern University, Khoury College of Computer Sciences 01/2026 - 05/2028 (Expected)
Master of Science in Computer Science | GPA: 4.0 / 4.0
Coursework: Algorithms, Programming Design Paradigm; in progress: Machine Learning, NLP

Conestoga College, Graduate Certificate in Mobile Solutions Development, GPA 3.74 / 4.0 2022 - 2023
Coursework: Android and iOS Application Development, Android Wear, Mobile Capstone

Victoria University, Bachelor of Information Technology, GPA 6.25 / 7.0 2018 - 2021
Major: Web and Mobile Application Development

EXPERIENCE

Virtusa, Associate Engineer (Quality Assurance), Colombo, Sri Lanka 06/2021 - 03/2022

- Verified a British Telecommunications (BT) platform serving **1M+ users** by writing **Java and Selenium WebDriver** tests and running test plans in agile sprints.
- Closed about **10 tickets per 2-week sprint, 2-3 days early**, while handling peer reviews, client demos and meetings, and tracking defect density and open defect counts.

Food Services & Hospitality, Team Member, Waterloo, Canada 01/2024 - 12/2025

PROJECTS

Distributed Task Scheduler Case Study github.com/PubuduGunasekara/distributed-task-scheduler
Java 21, Spring Boot, Kafka, Redis, PostgreSQL, Prometheus, Docker, GitHub Actions

- Prevented duplicate task runs under Kafka's at-least-once delivery (**zero duplicates** in a 1,000-task load test) by combining a database state machine, optimistic locking and a Redis lock, with backoff retries, a dead-letter topic and a recovery job.
- Sustained **~2,700 tasks/min** (3 consumers, local) with **188 tests, 93.4% line and 91.1% branch coverage** (80% CI gate), on a hexagonal architecture enforced by ArchUnit.

ResumeMatch github.com/PubuduGunasekara/rag-resumematch
Python, FastAPI, ChromaDB, Ollama, Next.js, Docker, GitHub Actions

- Matched a resume against **107,528 LinkedIn postings** (237,902 chunks) in **13.8 s** median, fully on a laptop, by building a **RAG** pipeline: bge-small embeddings, ChromaDB search with rank fusion and a 3B model via Ollama.
- Caught unsupported **RAG** claims (the model cited the wrong posting for **73%** of its “matched” requirements, 35 of 48) by checking every claim in code before it is shown, and raised recall@30 from **0.79 to 0.97** by tuning the vector index. **226 tests**, CI.

AI Code Review Assistant Live Demo github.com/PubuduGunasekara/ai-code-reviewer
React 19/Vite, Node.js/Express, PostgreSQL, Redis, gpt-4o-mini, Docker, AWS

- Delivered pull-request reviews in **8.4 s at ~$0.002 each** (5 test reviews) with a GitHub OAuth app (React on AWS Amplify, Express on EC2) that returns a severity, category, CWE reference and fix per finding.
- Made model output dependable with strict JSON-schema output (OpenAI Structured Outputs), so a malformed response fails the review, plus an hourly per-user rate limit (Lua) and Redis cache. 19 tests.

Smart Farm Demo github.com/PubuduGunasekara/smart-farm-1.1.0
React Native (Expo), Redux, Firebase, Arduino, ESP

- Built the device-control section of a team IoT app and the physical Arduino farm model, sending water, feed, gate and cleaning commands over Firebase to WiFi Arduino/ESP controllers that drive relays and motors.
- **Top 10 Most Innovative Projects**, NSBM Green EXE v1.0 software competition (team project).

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL, C++ (Arduino)
Backend & AI: Spring Boot, Node.js/Express, FastAPI, REST APIs, Kafka, Redis, PostgreSQL, RAG, OpenAI API
Infra & testing: Docker, AWS, Linux, GitHub Actions, CI/CD, Prometheus, Selenium
Frontend & mobile: React, Next.js, React Native (Expo), Redux, Firebase
Certifications: Azure Fundamentals (AZ-900), IBM Full Stack Software Developer, Meta Android Developer